

Nwamadi Ikechukwu Michael

Data Engineer

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SUMMARY

Results-driven **Data Engineer with 3+ years of experience** designing, building, and optimizing large-scale data pipelines and analytics solutions across diverse industries. Proven expertise in **ETL development, data modeling, cloud platforms (AWS, Azure, GCP), distributed systems, and big data frameworks**. Skilled in **Python, SQL, Apache Spark, Airflow, Kafka, Snowflake, and CI/CD pipelines**, delivering **scalable, secure, and business-impactful data solutions**. Adept at collaborating with **cross-functional teams** to translate business needs into **high-quality, production-grade data products**. Authorized to work in the U.S. (EAD – No sponsorship required).

PROFESSIONAL EXPERIENCE

Data Engineer | McKinsey & Company, MD, USA

Oct 2023 – Current

- Designed **scalable, high-performance ETL pipelines** using **PySpark, AWS Glue, and Apache Airflow**, enabling automated ingestion, transformation, and loading of structured/unstructured data from multiple sources, reducing overall processing time by 35% and supporting near-real-time analytics.
- Built and maintained an **enterprise-grade data lake architecture** on AWS S3 with optimized partitioning strategies, lifecycle policies, and compression formats (Parquet, ORC), achieving a 20% reduction in storage costs while ensuring faster query execution.
- Developed **dimensional data models** (Star/Snowflake schemas) and materialized views in **Snowflake** to support BI and analytics teams, improving query performance by 40% and enabling self-service reporting capabilities.
- Integrated **real-time streaming data pipelines** from Apache Kafka and AWS Kinesis into Amazon Redshift, implementing transformation logic for event-based processing, reducing operational reporting delays and enabling instant KPI tracking.
- Created and deployed **automated data quality frameworks** in Python with AWS Lambda triggers to perform schema validation, null checks, duplicate detection, and business rule validations, ensuring 99% accuracy across enterprise reports.
- Implemented **CI/CD deployment pipelines** using Jenkins and Git for ETL jobs, Airflow DAGs, and Snowflake SQL scripts, reducing deployment times from hours to under 30 minutes and ensuring zero downtime releases.
- Collaborated with **data science teams** to design reusable **feature engineering pipelines** for machine learning models, incorporating advanced aggregations and time-series transformations, resulting in a 15% improvement in model prediction accuracy.
- Applied **data governance and compliance measures** including GDPR, HIPAA, and encryption-at-rest/in-transit (KMS, SSL), implementing RBAC in Snowflake to protect sensitive data and pass all security audits.
- Partnered with stakeholders to **define KPIs and deliver interactive BI dashboards** in Power BI and Tableau, directly connected to Snowflake datasets, reducing ad-hoc report requests by 50% and improving decision-making speed.
- Operated in an **Agile/Scrum environment**, leading sprint planning, backlog grooming, daily stand-ups, and retrospectives to ensure timely delivery of high-priority data engineering projects.

Data Engineer | BCBS, MD, USA

Aug 2021 – Sep 2023

- Designed and deployed **large-scale data ingestion frameworks** using Apache Spark, Hadoop, and Hive to process 10TB+ of structured, semi-structured, and unstructured datasets daily, improving data pipeline efficiency and scalability.
- Led the **migration of an on-premises Oracle data warehouse to Azure Synapse Analytics**, building automated ETL processes in Azure Data Factory and achieving a 25% reduction in infrastructure costs while improving query performance.
- Built **incremental and real-time ETL workflows** in Azure Data Factory to process streaming data feeds with a latency of under 2 minutes, enabling near-real-time operational dashboards.
- Developed Python-based **metadata management and lineage tracking automation scripts**, integrating with Azure Purview for improved data governance, regulatory compliance, and faster issue resolution.
- Created **OLAP cubes and optimized dimensional data models** to support high-speed BI queries in tools like Power BI and Tableau, reducing report generation time by 30% and improving user adoption.
- Integrated external data sources through **third-party APIs**, performing enrichment and harmonization with existing enterprise datasets to enhance the accuracy and completeness of advanced analytics initiatives.
- Performed **Hadoop cluster optimization and Spark job tuning**, improving data processing performance by 20% through executor tuning, shuffle optimization, and memory configuration enhancements.
- Developed **role-based access controls (RBAC)** and encryption policies for sensitive datasets in Hive and Azure Synapse, ensuring compliance with corporate data security standards and industry regulations.
- Worked with QA and UAT teams to **validate ETL output accuracy** through test cases, data reconciliation, and defect resolution, reducing post-deployment data issues by 90%.
- Mentored junior data engineers on **SQL query optimization, PySpark transformations, and data modeling best practices**, improving overall team productivity and reducing onboarding time for new hires.

TECHNICAL SKILLS

- **Data Analysis & Visualization:** Python, SQL, Java, Scala, PySpark, C Python (NumPy, Pandas), Shell Scripting, R
- **Data Engineering:** Python, SQL, MongoDB, Grok, APIs, Regex, JSON, YAML, Azure Data Factory (ADF), Databricks, AWS, Azure DevOps, SFTP, SDLC, Docker, Kubernetes, CloudWatch
- **Big Data & Machine Learning:** Apache Spark, Hadoop, Hive, Scikit-Learn, TensorFlow, Keras, Hadoop, HDFS, Scala, SAP HANA
- **Cloud Platforms:** AWS, AWS GLUE, AWS Redshift, AWS Lambda, Google Cloud Azure Databricks, Data Lake (ADLS), Cosmos DB, AWS S3, EMR, DynamoDB
- **Data Warehousing & ETL:** ETL Development, Data Modeling, Data Lakes, Amazon Redshift, Google Big Query, Azure Data Factory, Azure Databricks, Azure Synapse, Snowflake
- **Data Visualization & BI Tools:** Tableau, Power BI, Microsoft Office Suite, Power Pivot, Power View, Power Query
- **Databases:** MySQL, PostgreSQL, NoSQL (MongoDB, Cassandra), Oracle, Teradata, SQL Server, PL/SQL, BigQuery □ **Version Control & CI/CD:** Git, Jenkins, Docker
- **Data Management:** Data Migration, Data Cleansing, Data Integration, SQL, ETL Processes
- **Tools & Technologies:** Microsoft Visio, Tableau, Power BI, MuleSoft, Excel (Advanced), SAP BI, SQL, Tableau, Power BI
- **Project Management:** Agile methodologies (Scrum, Kanban), Jira, Confluence, ServiceNow, Sprint Planning, Backlog Management

EDUCATION

Master's Degree | Morgan State University, Baltimore, MD, USA

May 2021